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Over-sampling ensemble methods for class imbalanced datasets

Min-Wei Huang
School of Medicine, Kaohsiung Medical University, Kaohsiung, Taiwan; Kaohsiung Municipal Kai-Suan Psychiatric Hospital, Kaohsiung, Taiwan and Department of Physical Therapy, College of Health Care, Graduate Institute of Rehabilitation Science, China Medical University, Taichung, Taiwan
Chih-Fong Tsai and Hsin-Yi Lin
Department of Information Management, National Central University, Taoyuan, Taiwan, and
Wei-Chao Lin
Department of Information Management, Chang Gung University, Taoyuan, Taiwan; Department of Digital Financial Technology, Chang Gung University, Taoyuan, Taiwan and Department of Emergency Medicine, Chang Gung Memorial Hospital at Linkou, Taoyuan, Taiwan

Abstract

Purpose – One widely adopted approach for effectively handling class-imbalanced datasets is data over-sampling, which involves generating synthetic samples for the minority class. Among these methods, the synthetic minority oversampling technique (SMOTE) is one of the most commonly used algorithms across various domain-specific imbalanced datasets. Numerous SMOTE variants have been proposed in the literature to enhance performance. However, no single algorithm consistently outperforms others across all types of domain-specific datasets.

Design/methodology/approach – This paper introduces several ensemble over-sampling methods, which combine the results of multiple over-sampling algorithms applied in both parallel and sequential manners. In the experiments, 58 binary-class datasets are used in Study One, while Study Two involves 10 datasets from various medical domains, encompassing both binary and multi-class classification tasks. Additionally, four baseline over-sampling algorithms, i.e. SMOTE, Poly-Fit-SMOTE, ProWSyn and SMOTE-IPF, are evaluated alongside three classifiers: C4.5, SVM and XGBoost.

Findings – The experimental results demonstrate that the parallel-based ensemble method, which combines ProWSyn and SMOTE-IPF and selects the k nearest synthetic samples around the corresponding minority class centers, yields the best performance. Using this ensemble approach, the XGBoost classifier achieves superior AUC and G-mean results compared to those obtained with the four baseline over-sampling algorithms and most other ensemble methods.

Originality/value – This paper presents parallel and serial ensemble methods that integrate multiple over-sampling results to enhance the performance of individual over-sampling algorithms. The most effective ensemble method and its associated combined algorithms can serve as a representative baseline for future research in class imbalance learning.

Keywords Data science, Machine learning, Class imbalance learning, Over-sampling, Ensemble learning

Paper type Research article

1. Introduction

Class imbalance learning is a critical research challenge in machine learning, as many real-world datasets exhibit class imbalance. Examples include financial distress prediction (Shen

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et al., 2021; Sun *et al.*, 2020), fraud detection (Sisodia and Sisodia, 2020, 2022a), intrusion detection (Bagui and Li, 2021), cancer prediction (Fotouhi *et al.*, 2019), software defect prediction (Pandey and Tripathi, 2021) and fake news detection (Budhi *et al.*, 2021). In general, class-imbalanced datasets possess characteristics – such as small sample sizes, class overlap, and small disjuncts – that hinder the effectiveness of most conventional machine learning techniques. Consequently, classifiers often yield biased results (Branco *et al.*, 2016; Lopez *et al.*, 2013). The severe imbalance in sample distribution between classes in the training set makes it particularly difficult to distinguish minority class instances from majority class ones in the feature space.

To address this issue, data level approaches are commonly adopted. These methods aim to rebalance class distributions through re-sampling techniques, primarily categorized into under-sampling and over-sampling methods. Under-sampling reduces the number of majority class samples, while over-sampling generates synthetic minority class samples (Chen *et al.*, 2024; Galar *et al.*, 2012; He and Garcia, 2009; Rezvani and Wang, 2023; Sisodia and Sisodia, 2021).

In under-sampling, random selection is a widely used technique that removes majority class examples at random. It is often combined with ensemble classifiers by constructing multiple classifiers from different sampled subsets, then aggregating their predictions. Another approach involves clustering the majority class samples and removing unrepresentative instances based on distances from cluster centers, thereby reducing class size (Hoyos-Osorio *et al.*, 2021; Lin *et al.*, 2017; Tsai *et al.*, 2019).

On the other hand, over-sampling methods focus on generating new synthetic samples for the minority class. Among them, the synthetic minority oversampling technique (SMOTE) is particularly popular, as it interpolates between neighboring minority class instances to create new samples (Azhar *et al.*, 2023; Fernandez *et al.*, 2018; Elreedy *et al.*, 2024). Numerous SMOTE variants have been proposed to improve performance, including borderline-SMOTE (Han *et al.*, 2005), SMOTE-IPF (Saez *et al.*, 2015), Poly-Fit-SMOTE (Gazzah and Amara, 2008), G-SMOTE (Sandhan and Choi, 2014), LVQ-SMOTE (Nakamura *et al.*, 2013), and assembled SMOTE (Zhou *et al.*, 2013). Kovacs (2019) conducted an empirical comparison of 85 over-sampling methods on 104 imbalanced datasets, demonstrating that some SMOTE variants consistently outperform the original SMOTE.

However, different over-sampling methods often generate distinct synthetic samples even when applied to the same dataset. To the best of our knowledge, there has been limited research on integrating the outputs of multiple over-sampling algorithms.

To address this gap, ensemble learning can be employed. This approach constructs multiple classifiers and combines their predictions, typically achieving better performance than individual classifiers (Amuru *et al.*, 2025; Mo *et al.*, 2024; Rahman *et al.*, 2024). In related work, ensemble learning has also been applied to feature selection, referred to as ensemble feature selection, where multiple algorithms select features and the results are aggregated into a final subset (Ding *et al.*, 2025; Lohrmann *et al.*, 2025; Sisodia and Sisodia, 2022b; Zheng *et al.*, 2024).

This study extends the ensemble concept to the data-level re-sampling stage by proposing methods that combine synthetic datasets generated by different over-sampling algorithms. Two combination strategies, parallel and serial, are introduced (see Section 3).

This paper makes two primary contributions:

(1) It proposes novel parallel and serial over-sampling ensemble approaches that combine multiple over-sampling algorithms. These methods demonstrate how ensemble learning can be effectively applied at the data level to address class imbalance.

(2) Through two experimental studies, one involving 58 binary-class datasets and another comprising 10 datasets from medical domains with binary and multi-class settings, this paper identifies the most effective ensemble strategy. The resulting approach can serve as a baseline method for future research on class imbalance learning.

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The organization of this paper is as follows. Section 2 gives an overview of the over-sampling and related algorithms used in this paper. In addition, a short description of ensemble learning techniques is provided. Section 3 introduces the over-sampling ensemble methods including the parallel and serial combination approaches. Section 4 presents the experimental setup and the results. Finally, Section 5 concludes the paper.

2. Literature review

2.1 Over-sampling techniques

The goal of over-sampling techniques is to generate additional training samples for the minority class in order to rebalance originally class-imbalanced datasets. As noted by Fernandez *et al.* (2018), over-sampling has been widely adopted and proven effective across various domains. One key advantage is that over-sampling is performed during the data preprocessing stage, independently of the classifier construction process, allowing it to directly address the issue of insufficient minority class data. This study focuses on four well-known over-sampling algorithms, including the top three performers among the 85 methods evaluated in Kovacs (2019), as detailed below.

2.1.1 SMOTE. The SMOTE preprocessing method is widely regarded as the *de facto* standard for rebalancing class-imbalanced datasets (Fernandez *et al.*, 2018). In SMOTE, synthetic minority class samples are generated by interpolating between existing minority instances that satisfy a predefined neighborhood criterion. Specifically, the algorithm selects k nearest neighbors – typically with k set to 5 by default – for each randomly chosen minority instance. Synthetic samples are then created along the line segments connecting the selected instance to its nearest neighbors (Elreedy and Atiya, 2019).

In the literature, SMOTE has been extensively adopted and is often treated as the baseline over-sampling algorithm (Bej *et al.*, 2021; Shen *et al.*, 2021; Xie *et al.*, 2022; Zhu *et al.*, 2019). It is also commonly used in conjunction with other learning approaches, such as cost-sensitive classification techniques (Raghuwanshi and Shukla, 2021) and ensemble classification methods (Mahadevan and Arock, 2021; Sun *et al.*, 2020).

2.1.2 SMOTE-IPF. SMOTE-IPF, an extension of SMOTE, was developed to address noisy and borderline instances commonly found in class-imbalanced datasets. This method builds upon the iterative partitioning filter (IPF) approach and aims to not only rebalance the dataset but also remove noisy and borderline examples from the re-sampled data (Saez *et al.*, 2015).

The process involves two main steps. First, the dataset is rebalanced using SMOTE or other over-sampling techniques to adjust the class distribution. Second, IPF is applied to eliminate noisy samples, both from the original dataset and those generated during the over-sampling process. Specifically, IPF performs multiple iterations to identify and remove noisy instances, and the procedure terminates once the number of detected noisy examples falls below a predefined threshold relative to the size of the original training set.

2.1.3 Poly-Fit-SMOTE. In the study by Gazzah and Amara (2008), synthetic data samples were generated using polynomial fitting functions. Four topological approaches were proposed for this purpose: star topology, bus topology, polynomial curve topology and mesh topology. Among these, the polynomial curve topology demonstrated superior performance, particularly when the over-sampling ratio was set to 1.

In Poly-Fit-SMOTE, the optimal trend curve for a set of minority class instances is estimated by computing the coefficients of a polynomial function p(x)p(x)p(x) of degree nnn. This polynomial is fitted to the feature space using the *least squares error* method by

$$p(x) = p_1^{x^n} + p_2^{x^{n-1}} + p_3^{x^{n-2}} + \ldots + p_n^{x^n} + p_{n+1}. \tag{1}$$

Then, k linearly spaced values x_k ($k \in [-1, 1]$) are generated by over-sampling, followed by the creation of new synthetic minority examples based on evaluating the value of the polynomial $p(x)$ at x_k.

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2.1.4 ProWSyn. The Proximity Weighted Synthetic Oversampling Technique (ProWSyn) generates synthetic samples for the minority class by assigning weight values based on proximity information. Specifically, the distance of each minority instance from the decision boundary guides the distribution of the generated synthetic samples across the minority class space (Barua *et al.*, 2013).

The method involves three main steps. First, the number of synthetic samples required for the minority class is determined. Second, each minority instance x is assigned a weight based on its proximity level, which is calculated using the Euclidean distance from the majority class instances. Finally, synthetic samples are generated according to these weighted proximities, resulting in an oversampled dataset that better represents the minority class distribution.

2.2 Ensemble learning techniques

2.2.1 The concept. Ensemble learning was developed to enhance the classification performance of individual classifiers. The fundamental idea is to combine multiple complementary classifiers to compensate for the errors made by individual models (Frosyniotis *et al.*, 2003; Ho *et al.*, 1994).

The most straightforward approach for combining the outputs of multiple classifiers is majority voting, where the class label that receives the highest number of votes is selected as the final prediction of the ensemble (Kittler *et al.*, 1998).

Two of the most widely used ensemble techniques are bagging and boosting. In bagging (bootstrap aggregating), multiple classifiers are trained independently using different subsets of the training data. These subsets are typically generated using the bootstrap method, which creates k replicated training sets by sampling with replacement from the original dataset (Breiman, 1996).

In contrast, boosting builds classifiers sequentially. Initially, weights are assigned to each training instance. After training the k-th classifier, it is evaluated on the original training set to identify misclassified instances. In the next iteration, the $(k+1)$-th classifier is trained using a modified dataset in which higher weights are given to the previously misclassified instances. This iterative process continues until K classifiers are trained, each focusing increasingly on the more difficult examples (Freund and Schapire, 1996).

2.2.2 Ensemble learning in class imbalanced datasets. Ensemble learning techniques are frequently employed to address the challenges posed by class-imbalanced datasets. Specifically, under-sampling and over-sampling methods are often combined with ensemble classifiers constructed using bagging or boosting strategies. For example, SMOTEBoost integrates SMOTE to rebalance the training set during each boosting iteration, thereby enabling the construction of multiple classifiers (Chawl *et al.*, 2003). In RUSBoost, SMOTE is replaced with a random under-sampling method to create balanced datasets for boosting-based classifier training (Seiffert *et al.*, 2010).

In bagging-based approaches, random under-sampling is repeatedly applied to the original training set, producing multiple diverse subsets – each used to train a different classifier (Liu *et al.*, 2008). Similarly, SMOTEBagging employs SMOTE to generate synthetic minority samples, and then uses bootstrap sampling to create multiple training sets for bagging-based ensemble construction (Wang and Yao, 2009).

However, different re-sampling techniques typically produce distinct balanced training sets, even when applied to the same imbalanced dataset. As a result, traditional ensemble approaches generally rely on only one specific under- or over-sampling method to rebalance the dataset before constructing the ensemble.

In this study, we explore the possibility of applying multiple over-sampling algorithms simultaneously to a given training set. The resulting balanced datasets are then systematically combined into a unified dataset for classifier construction. In essence, the objective is to investigate whether an ensemble of over-sampling methods, referred to as over-sampling

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ensembles, can produce a more effective and better-balanced training set compared to using a single over-sampling method.

3. Over-sampling ensemble methods

Two approaches for combining the multiple balanced training sets generated by different over-sampling algorithms are introduced in the following sections: the parallel combination approach and the serial combination approach.

3.1 Parallel combination approaches

In parallel combination approaches, multiple over-sampling algorithms are applied simultaneously. In this study, we consider combinations involving two or three over-sampling algorithms to avoid excessive computational cost.

For example, consider combining three over-sampling algorithms, denoted as *A*, *B* and *C*, on a given training set *D*, which contains 100 majority and 10 minority class instances. The goal is to generate 90 synthetic minority class instances to balance the dataset. Each over-sampling algorithm is independently applied to produce 90 synthetic samples, resulting in three balanced training sets: D_A, D_B, and D_C. Each of these sets contains 100 majority and 100 minority instances. For instance, D_A is composed of the original 10 minority instances and 90 synthetic samples generated by algorithm *A*. Note that the 90 synthetic instances in D_A, D_B, and D_C are not necessarily identical.

The next step involves selecting subsets from D_A, D_B, and D_C. The proportion of selected synthetic samples can be equal (e.g. 30 samples from each set) or weighted (e.g. 27 from D_A, 27 from D_B and 36 from D_C. The choice of proportion will be discussed further in Section 4.1.2.

Once the proportions are defined, we need a mechanism to select specific synthetic instances. The four proposed selection strategies are described below.

3.1.1 The random method. The random method simply selects the required number of synthetic instances from each set at random. For example, with a 30%: 30%: 40% split, 27, 27 and 36 synthetic samples are randomly selected from D_A, D_B and D_C, respectively, resulting in subsets, S_A, S_B and S_C. These subsets are then combined with the 100 majority class examples from *D*, forming a new balanced datasets with 100 majority and 100 minority instances.

3.1.2 The center method. The center method selects synthetic instances closest to the centroid of the minority class in each data. For instance, to select 27 samples (30%) from D_A, the algorithm computes the centroid of the 90 synthetic samples generated by *A* and selects the 27 closest instances. The same procedure is applied to D_B, and D_C, resulting in S_B and S_C. These subsets are then merged with the majority class data to form the final training set.

3.1.3 The cluster random method. In the cluster random method, *k*-means clustering is performed on the synthetic samples in each dataset. The number of clusters is determined using the elbow method (Liu and Deng, 2021). Let $C_{A,i}$ denote the *i*-th cluster in dataset D_A. The percentage of synthetic samples in each cluster is calculated as follows:

$$Percent_{C_i} = \frac{|C_i|}{N}, \tag{2}$$

Where $|C_i|$ is the number of synthetic samples in cluster *i*, and *N* is the total number of synthetic samples.

Given a desired number *K* of samples to select, the number of selected from cluster *i* is as follows:

$$n_i = Percent_{C_i} \times K, \tag{3}$$

These samples are selected randomly within each cluster. The sample process is applied to D_B, and D_C, producing subsets S_A, S_B and S_C.

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For example, for D_A, there are three clusters identified, denoted as C_1, C_2 and C_3, which contain 20, 30 and 40 synthetic examples, respectively. The percentages of the synthetic examples for the three clusters are found to be 2/9, 3/9 and 4/9, respectively. Then, the number of synthetic examples randomly selected for each of the three clusters is calculated as follows:

$$6 = 2/9 \times 30\% \times 90 \text{ for } C_1; \tag{4}$$

$$9 = 3/9 \times 30\% \times 90 \text{ for } C_2; \tag{5}$$

$$12 = 4/9 \times 30\% \times 90 \text{ for } C_3. \tag{6}$$

Therefore, D_A, composed of 27 synthetic examples, is produced. Similarly, this procedure can be employed for D_B and D_C, individually.

3.1.4 The cluster center method. The cluster center method uses the same clustering procedure as the cluster random method but selects the k nearest samples to the cluster centroids instead of selecting randomly. For example, if C_1 in D_A contains 20 samples and contributes six samples, the six nearest samples to the centroid of C_1 are chosen. This process is repeated for each cluster across D_A, D_B and D_C, forming S_A, S_B and S_C, which are then combined with the majority class examples.

Figure 1 presents the pseudocode for the parallel combination approaches.

3.2 Serial combination approaches

In serial combination approaches, over-sampling algorithms are applied sequentially. As with parallel methods, we consider combinations of two and three over-sampling algorithms.

Let the original training set D contain 100 majority and 10 minority class samples. The goal is to generate 90 synthetic minority instances. For example, assume two algorithms, A and B, are applied in sequence with generation rates of 30% and 70%, respectively.

(1) Stage 1: algorithm A generates 27 synthetic samples (30% of 90), yielding 37 minority instances (10 original + 27 synthetic)

(2) Stage 2: algorithm B is applied to these 37 instances, generating 63 synthetic samples (70% of 90).

(3) Final output: 100 majority + (10 original + 27 + 63) = 100 minority class instances.

Note the ratio setting for these serial combinations are detailed in Section 4.1.2.

Figure 2 presents the pseudo for the serial combination approaches.

4. Experiments

4.1 Experimental setup

4.1.1 Datasets. The datasets used in the experiments were collected from the KEEL dataset repository [1], which contains 106 binary class datasets. The "easy" datasets with classification accuracies of higher than 0.9 were excluded. This left 58 datasets for use in subsequent experiments, with class imbalance ratios ranging from 1.82 to 129.44, numbers of features from 3 to 18 and numbers of data samples from 101 to 4,174. The relevant information for these datasets is shown in Table 1.

Each dataset was divided into 80% training and 20% testing sets based on five-fold cross validation. In addition, the class imbalance ratio of the training set in each fold is similar to that of the original dataset, to avoid producing biased results.

4.1.2 Over-sampling algorithms. The four single over-sampling algorithms employed in the ensembles included SMOTE, SMOTE-IPF, Poly-Fit-SMOTE, and ProWSyn. The three algorithm combinations are based on SMOTE-IPF, Poly-Fit-SMOTE and ProWSyn. The top

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```
Let ID = input data
Let the data which is generated by poly-nominal = poly
Let the data which is generated by Prownsyn = prow
Let the data which is generated by SMOTE-IPF = ipf
Choose the generated data to reach equal numbers between the majority and minority
class
if combine two techniques (EO2):
   ensemble ratio = [[5,5,0],[2,8,0],[3,7,0]]
else if combine three techniques (EO3):
    ensemble ratio = [[4,4,2],[3,3,4],[2,5,3]]
Use the following 4 methods separately to choose the data based on the above ratio
(1) Random method: randomly choose the data depends on the ratio
(2) Center method:
    Step1: Calculate the center of the generated data
    Step2: Choose the data which is close to the center
(3) Cluster random method:
    Step1: Use elbow method to calculate the optimal number of clusters
    Step2: Calculate the number Dn which has to be chosen according to the
    ratio between each cluster and the whole data numbers
                              Dn = ratio × (Cn/N)
    Let ratio = the assigned ratio for the technique
    Let Cn = the numbers of per cluster
    Let N = the numbers of whole data
    Step3: Choose the data randomly in each cluster
(4) Cluster center method:
    Step1: Use elbow method to calculate the optimal number of clusters
    Step2: Calculate the center of each cluster
    Step3: Calculate the number which has to be chosen according to the ratio
    between each cluster and the whole data numbers
    Step4: Choose the data which is close to the center of each cluster
Let the choosing data = CD
Let the number of choosing data = N
for each method:
    for I in Permutate ensemble ratio:
        Cdpoly = poly × 0.1 × i[0] × N
          Cdprowsyn = prowsyn × 0.1 × i[1] × N
          Cdipf = ipf × 0.1 × i[2] × N
   end for
end for
temp = concat(Cdpoly, Cdprowsyn, Cdipf)
EOResult = Merge(temp, OD)
```

Figure 1. The pseudocode for the parallel combination approaches

two of these three algorithms are selected for the two-over-sampling algorithm combination. The relevant parameters for these four algorithms are listed in Table 2.

To implement the parallel and serial combination approaches, it is necessary to determine appropriate proportions for generating synthetic data subsets using different over-sampling algorithms. For ensembles comprising three algorithms, the candidate proportion settings include 1:3:6, 2:2:6, 3:3:4, 4:4:2 and 2:5:3. In the case of two-algorithm ensembles, the considered proportions are 1:9, 2:8, 3:7, 4:6 and 5:5. In total, ten distinct proportion settings are evaluated for both the parallel and serial combination approaches.

To reduce the computational burden of conducting large-scale experiments on all 58 datasets, we sought to identify a subset of the most effective proportion settings. An initial pilot study was conducted for this purpose, in which three representative datasets were selected to

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```
Let ID = input data
Let tech = [poly, Prownsyn, IPF]
Choose the generated data to reach equal numbers between the majority and minority class
if combine two techniques (EO2):
    ensemble ratio = [[5,5,0],[2,8,0],[3,7,0]]
else if combine three techniques (EO3):
     ensemble ratio = [[4,4,2],[3,3,4],[2,5,3]]
for m in ensemble ratio:
    for i in Permutate (m) ratio:
     PolyData = Poly.sample(ID, propagation=i[0])
     PrownsynData = Prownsyn.sample(polydata, propagation = i[1])
     IPFData = SMOTE_IPF.sample(PrownsynData, propagation = i[2])
    end for
end for
EOResult = Merge(IPFData, ID)
```

Figure 2. The pseudocode for the serial combination approaches

evaluate the performance of the ten candidate proportion settings. The selection of representative datasets was based on k-means clustering ($k = 3$) applied to the 58 datasets, using input features such as class imbalance ratios, the number of features and the number of instances. The datasets closest to each cluster centroid were selected, resulting in the choice of yeast-1_vs_7, yeast6 and abalone19.

Subsequently, the C4.5 classifier was used to evaluate the performance of four ensemble strategies, namely, Random, Center, Cluster Random and Cluster Center, across the ten different proportion settings using the three representative datasets.

The results indicate that no particular proportion setting yields a statistically significant improvement in performance. Therefore, to streamline subsequent experiments, only three proportion settings were selected: 2:8, 3:7 and 5:5 for two-algorithm ensembles and 2:5:3, 3:3:4 and 4:4:2 for three-algorithm ensembles. Due to space constraints, only the best-performing results are reported in the paper.

A similar trend was observed in the serial combination approach, where combining two or three over-sampling algorithms with the C4.5 classifier yielded comparable performance across different proportion settings. Consequently, the final proportion settings adopted for the serial ensemble methods are identical to those used in the parallel ensemble approaches.

4.1.3 Classifier design. In this study, three classifiers, which are C4.5, Support vector machine (SVM) and XGBoost, are constructed for performance comparison. All classifiers are implemented using Python 3.7, with their respective default parameter settings. Specifically, the SVM employs a radial basis function (RBF) kernel with parameters set to gamma = "auto", degree = 3 and C = 1. For XGBoost, the booster type is set to "gbtree", verbosity to 1, and the maximum depth to 6.

The classification performance of these models is evaluated using the area under the receiver operating characteristic curve (AUC), which is widely regarded as a robust and reliable metric for assessing classifiers on imbalanced datasets (Branco *et al.*, 2016; Elreedy and Atiya, 2019; Galar *et al.*, 2012; Kovacs, 2019).

4.2 Results on parallel combination approaches

Figures 3–5 present the average performance of the C4.5, SVM and XGBoost classifiers under various methods. It is important to note that the baseline refers to the classifiers trained on the original training set without any over-sampling applied. In the figure legends, EO2 denotes the ensemble of two over-sampling algorithms and EO3 refers to the ensemble of three over-sampling algorithms. The abbreviations R, C, CR, and CC represent the Random, Center, Cluster Random and Cluster Center ensemble strategies, respectively.

The results indicate that classifier performance improves over the baseline when the training set is rebalanced using over-sampling algorithms such as SMOTE, Poly-Fit-SMOTE,

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Table 1. Information related to the 58 datasets

	Datasets	Imbalance ratios	No. of features	No. of instances
1	glass1	1.82	9	214
2	Pima	1.87	8	768
3	glass0	2.06	9	214
4	yeast1	2.46	8	1,484
5	haberman	2.78	3	306
6	vehicle1	2.9	18	846
7	vehicle3	2.99	18	846
8	ecoli1	3.36	7	336
9	ecoli2	5.46	7	336
10	glass6	6.38	9	214
11	yeast3	8.1	8	1,484
12	ecoli3	8.6	7	336
13	yeast-2_vs_4	9.08	8	514
14	ecoli-0-2-3-4_vs_5	9.1	7	202
15	yeast-0-3-5-9_vs_7–8	9.12	8	506
16	glass-0-1-5_vs_2	9.12	9	172
17	yeast-0-2-5-6_vs_3-7-8-9	9.14	8	1,004
18	yeast-0-2-5-7-9_vs_3-6-8	9.14	8	1,004
19	ecoli-0-2-6-7_vs_3–5	9.18	7	224
20	ecoli-0-3-4-7_vs_5–6	9.28	7	257
21	yeast-0-5-6-7-9_vs_4	9.35	8	528
22	glass-0-1-6_vs_2	10.29	9	192
23	glass-0-1-4-6_vs_2	11.06	9	205
24	glass2	11.59	9	214
25	cleveland-0_vs_4	12.62	13	177
26	ecoli-0-1-4-6_vs_5	13	6	280
27	yeast-1_vs_7	14.3	7	459
28	glass4	15.47	9	214
29	ecoli4	15.8	7	336
30	page-blocks-1-3_vs_4	15.86	10	472
31	abalone9-18	16.4	8	731
32	zoo-3	19.2	16	101
33	glass-0-1-6_vs_5	19.44	9	184
34	yeast-1-4-5-8_vs_7	22.1	8	693
35	glass5	22.78	9	214
36	yeast-2_vs_8	23.1	8	482
37	flare-F	23.79	11	1,066
38	yeast4	28.1	8	1,484
39	winequality-red-4	29.17	11	1,599
40	poker-9_vs_7	29.5	10	244
41	yeast-1-2-8-9_vs_7	30.57	8	947
42	winequality-white-9_vs_4	32.6	11	168
43	yeast5	32.73	8	1,484
44	winequality-red-8_vs_6	35.44	11	656
45	ecoli-0-1-3-7_vs_2–6	39.14	7	281
46	abalone-17_vs_7-8-9-10	39.31	8	2,338
47	abalone-21_vs_8	40.5	8	581
48	yeast6	41.4	8	1,484
49	winequality-white-3_vs_7	44	11	900
50	winequality-red-8_vs_6–7	46.5	11	855
51	abalone-19_vs_10–11–12–13	49.69	8	1,622
52	winequality-white-3-9_vs_5	58.28	11	1,482
53	poker-8-9_vs_6	58.4	10	1,485
54	winequality-red-3_vs_5	68.1	11	691

(*continued*)

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Table 1. Continued

	Datasets	Imbalance ratios	No. of features	No. of instances
55	abalone-20_vs_8-9-10	72.69	8	1,916
56	poker-8-9_vs_5	82	10	2,075
57	poker-8_vs_6	85.88	10	1,477
58	abalone19	129.44	8	4,174

Table 2. Parameters for the over-sampling algorithms

Algorithms	Parameters
SMOTE	k_neighbors = 2
SMOTE-IPF	proportion = 1.0; n_neighbors = 5; n_folds = 9, k = 3; p = 0.01; voting = majority; classifier = DecisionTreeClassifier; n_job = 1
Poly-Fit-SMOTE	proportion = 1.0; topology = star
ProWSyn	proportion = 1.0; n_neighbors = 5; L = 5; theta = 1.0; n_jobs = 1

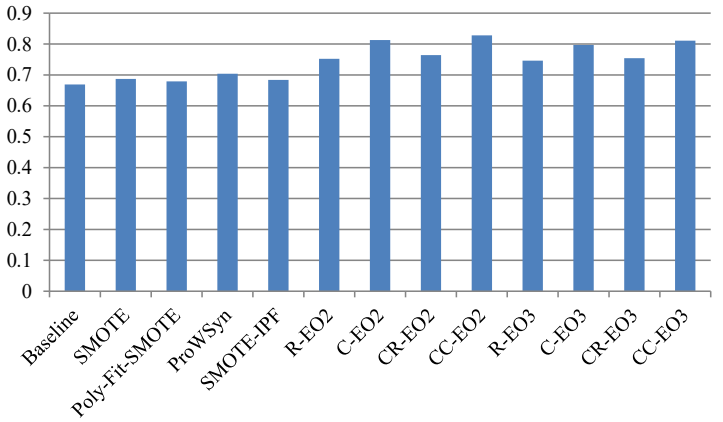


Figure 3. Average AUC rates of C4.5 obtained by different methods

ProWSyn and SMOTE-IPF. Among these, the ProWSyn algorithm consistently outperforms the other three.

Furthermore, regardless of the classifier employed, all ensemble methods demonstrate superior performance compared to the individual over-sampling algorithms. Among the three classifiers, XGBoost achieves the highest overall performance. Specifically, the top three ensemble methods for XGBoost are CC-EO2 (AUC = 0.838), C-EO2 (AUC = 0.836) and CC-EO3 (AUC = 0.834). For the C4.5 classifier, the corresponding top three methods are CC-EO2 (AUC = 0.828), C-EO2 (AUC = 0.813), and CC-EO3 (AUC = 0.811).

Figures 6–8 illustrate the performance of the C4.5, SVM and XGBoost classifiers under different methods based on the G-mean evaluation metric. These results are consistent with those obtained using AUC, further confirming that CC-EO2, C-EO2 and CC-EO3 are the most effective ensemble strategies for both the C4.5 and XGBoost classifiers.

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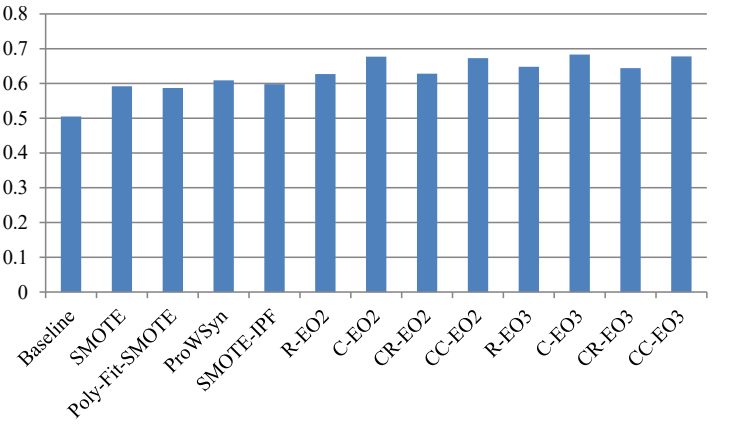


Figure 4. Average AUC rates of SVM obtained by different methods

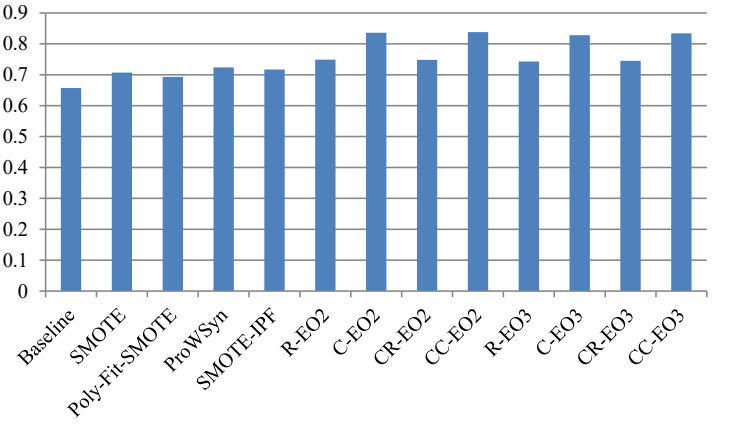


Figure 5. Average AUC rates of XGBoost obtained by different methods

The consistent results obtained from the XGBoost and C4.5 classifiers provide a reliable basis for identifying the most effective ensemble methods. Notably, there is no statistically significant difference in performance among the top three ensemble approaches, regardless of which classifier is used. This finding suggests that any of the top three ensemble methods can be confidently adopted in future applications aimed at enhancing the performance of XGBoost and C4.5 classifiers.

However, when computational efficiency is a critical consideration during the over-sampling stage, the C-EO2 method, constructed by combining the top two over-sampling algorithms, namely ProWSyn and SMOTE-IPF, is recommended due to its balance of performance and reduced computational cost.

4.3 Results of serial combination approaches

Figures 9–11 present the average AUC scores achieved by the C4.5, SVM and XGBoost classifiers across the different methods. The results indicate that there are no substantial performance differences between the serial EO2 and EO3 approaches for any of the classifiers.

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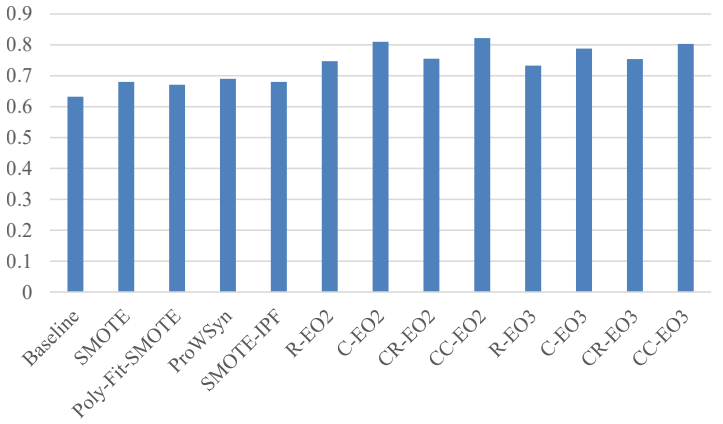


Figure 6. Average G-means of C4.5 obtained by different methods

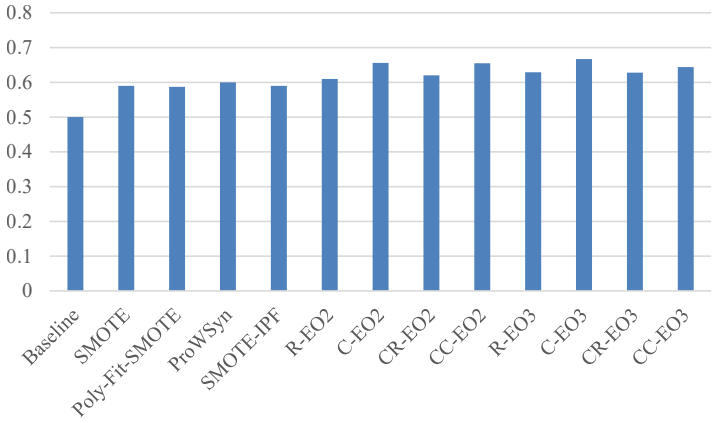


Figure 7. Average G-means of SVM obtained by different methods

Furthermore, when using SVM and XGBoost, the single best-performing over-sampling algorithm, ProWSyn, outperforms the serial combination methods, suggesting that the added complexity of serial combinations does not necessarily lead to improved performance in these cases.

On the other hand, Figures 12–14 display the G-mean results for the C4.5, SVM and XGBoost classifiers. The performances of the serial EO2 and EO3 methods are highly similar across all classifiers. Notably, the serial combination approaches lead to improved G-mean results only for the C4.5 classifier, outperforming the individual over-sampling algorithms. However, for SVM and XGBoost, the serial ensembles do not offer a significant advantage.

However, even when applying the best-performing serial combination method, EO2 combined with the C4.5 classifier, which yields an AUC of 0.723 and a G-mean of 0.720, the results remain noticeably inferior to those achieved by the top three parallel combination approaches using either XGBoost or C4.5. This finding suggests that serial over-sampling ensembles are suboptimal for handling class-imbalanced datasets and may not be recommended in practical applications.

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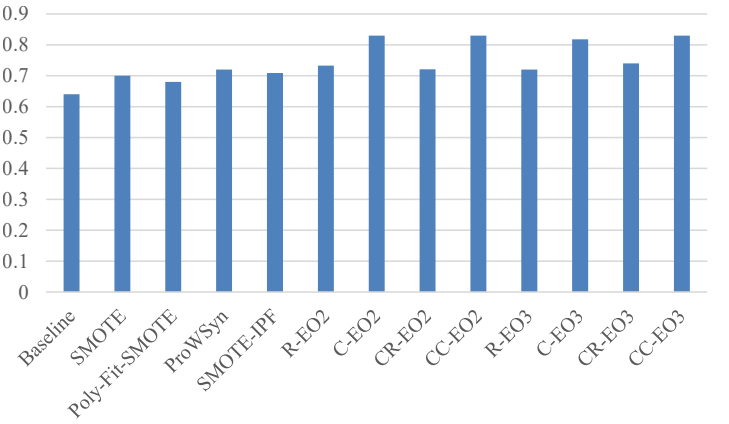


Figure 8. Average G-means of XGBoost obtained by different methods

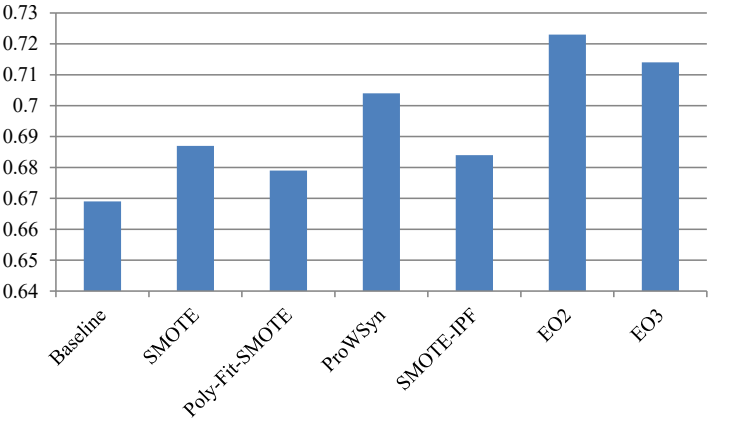


Figure 9. Average AUC rates of C4.5 obtained by different methods

4.4 Further comparisons

4.4.1 Results on medical domain problem datasets with binary and multiple classes. The effectiveness of the recommended over-sampling ensemble method, C-EO2, was further validated using 10 medical domain datasets, encompassing both binary and multi-class classification tasks, obtained from the OpenML repository [2]. Detailed information about these datasets is provided in Table 3. It is worth noting that the labels "B" and "M" in the "Classes" column indicate binary and multi-class datasets, respectively. A distinguishing characteristic of these datasets is their high dimensionality, which contrasts significantly with the 58 KEEL datasets used in earlier experiments.

Tables 4 and 5 present the AUC scores and G-mean values, respectively, obtained by the different methods using the XGBoost classifier. The best result for each dataset is highlighted in bold. On average, the C-EO2 method achieves the highest overall performance, while the four individual over-sampling algorithms consistently outperform the baseline. However, the performance differences among the four single methods are relatively minor. In summary, the proposed C-EO2 method demonstrates robust and consistent effectiveness, making it a

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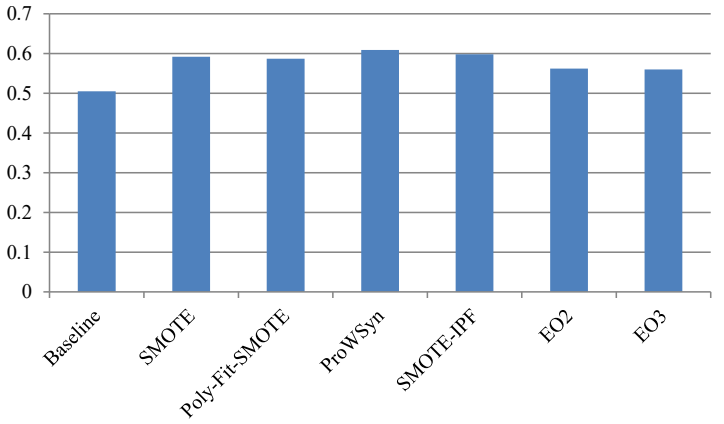


Figure 10. Average AUC rates of SVM obtained by different methods

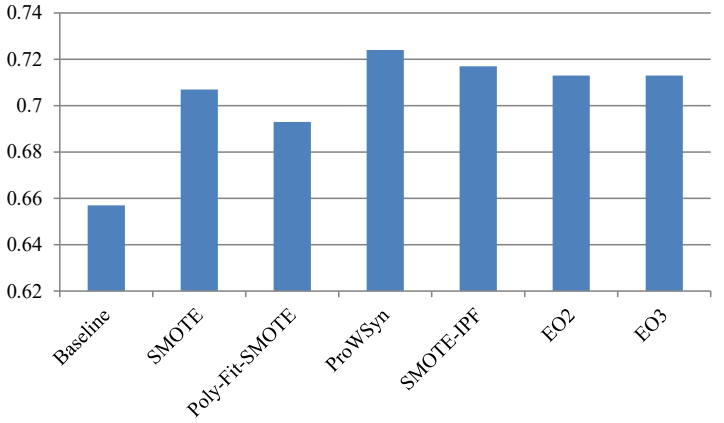


Figure 11. Average AUC rates of XGBoost obtained by different methods

strong candidate for a standard over-sampling strategy in class-imbalanced classification tasks.

4.4.2 Hyperparameter sensitivity analysis. To evaluate the robustness of XGBoost across different over-sampling methods, a hyperparameter tuning procedure was conducted. The hyperparameters were varied as follows: number of trees = {100, 400, 700, 1,000}, learning rate = {0.1, 0.2, 0.3}, and max depth = {3, 6, 10}. Figures 15 and 16 illustrate the average performance and corresponding standard deviations of XGBoost under each method.

The standard deviation results indicate that the XGBoost classifier remains highly robust across different hyperparameter settings, regardless of the over-sampling algorithm applied. In all cases, the performance variations remain within a narrow range, with differences below 0.03. Specifically, for the baseline XGBoost combined with over-sampling, the observed performance variations are 0.032 for AUC and 0.041 for G-mean.

Although the smallest performance variations are achieved using SMOTE with an AUC deviation of 0.016 and a G-mean deviation of 0.023, the proposed C-EO2 method also demonstrates stable results, yielding an AUC deviation of 0.018 and a G-mean deviation of

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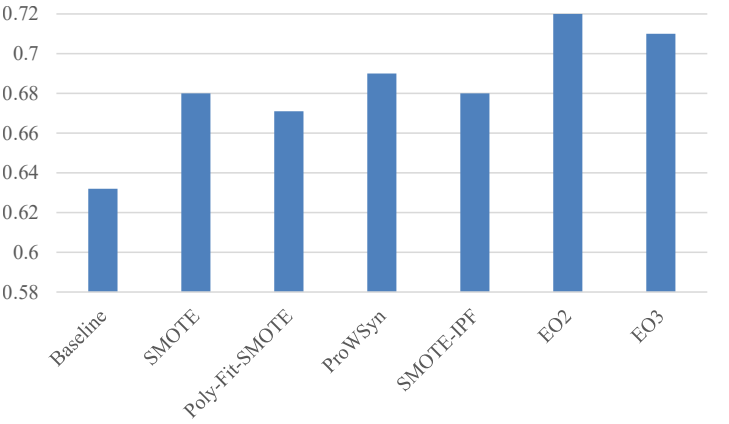


Figure 12. Average G-means of C4.5 obtained by different methods

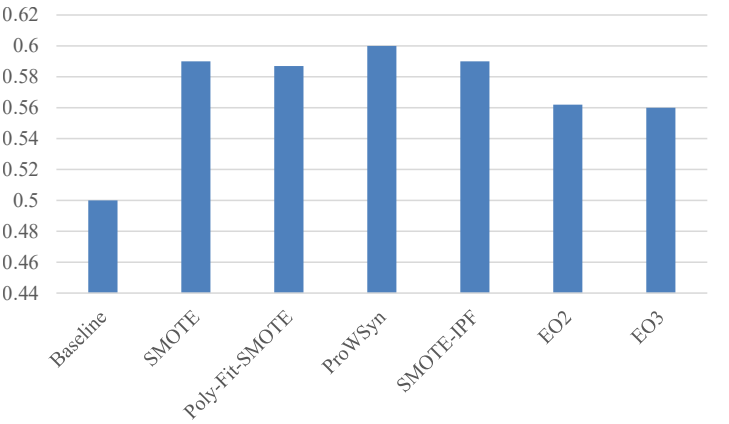


Figure 13. Average G-means of SVM obtained by different methods

0.030. These findings confirm the robustness of C-EO2 while maintaining competitive consistency across varying parameter configurations.

4.4.3 Feature distributions before and after over-sampling. To investigate the impact of over-sampling on model generalizability, we present a visual analysis using Multi-Dimensional Scaling (MDS) (Borg and Groenen, 2005) to illustrate feature distributions before and after over-sampling. The OVA_Uterus dataset is used as an example, and Figure 17 displays the feature distributions under the baseline, SMOTE and C-EO2 methods. Since the feature distributions resulting from Poly-Fit-SMOTE, ProWSyn and SMOTE-IPF are highly similar to that of SMOTE, they are excluded from this analysis for brevity. In the visualizations, the terms "positive" and "negative" refer to the minority and majority classes, respectively.

As shown in the baseline method (i.e. before over-sampling), the two classes are highly intertwined, making them difficult to distinguish. In contrast, after applying SMOTE and C-EO2, the class distributions become more balanced in size. Although some overlap between the two classes remains, the underlying structure becomes more discernible. Notably, in the circled regions of Figure 17(b) and 17(c), the feature overlap is reduced more effectively by

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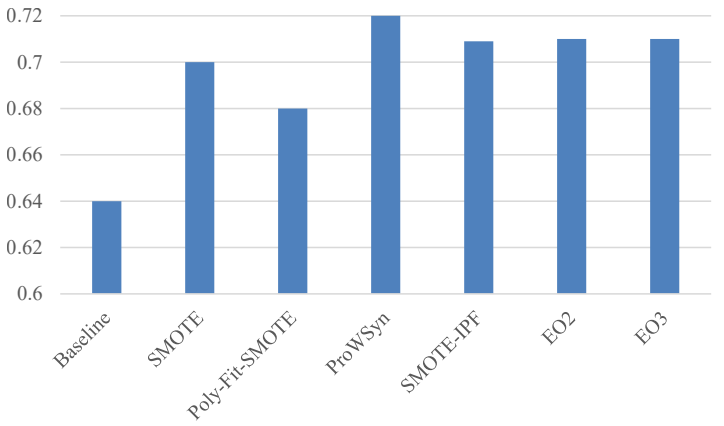


Figure 14. Average G-means of XGBoost obtained by different methods

Table 3. Dataset information

	Classes	No. of features	No. of instances	Imbalance ratios
BurkittLymphoma	M	22,284	220	2.9
GCM	M	16,064	190	3
hepatitisC	M	54,622	283	13.61
hiva_agnostic	B	1,618	4,229	27.38
lsvt	B	311	126	2
lung-multi	M	3,313	203	23.16
micro-mass	M	1,301	571	5.41
OVA_Ovary	B	10,937	1,545	6.8
OVA_Uterus	B	10,937	1,545	11.45
parkinson-speech-uci	B	754	756	2.93

Table 4. AUC rates of XGBoost obtained with different methods

	Baseline	SMOTE	Poly-Fit-SMOTE	ProWSyn	SMOTE-IPF	C-EO2
BurkittLymphoma	0.824	0.770	0.767	0.801	0.774	*0.849*
GCM	0.721	0.742	0.730	0.720	0.717	*0.747*
hepatitisC	0.644	*0.738*	0.725	0.683	0.649	0.736
hiva_agnostic	0.632	0.602	*0.658*	0.611	0.605	0.656
lsvt	0.662	0.715	*0.732*	0.683	0.701	0.705
lung-multi	0.816	0.873	0.831	0.868	0.867	*0.926*
micro-mass	0.867	0.862	0.870	0.879	*0.883*	0.875
OVA_Ovary	0.798	0.801	0.807	0.807	0.813	*0.830*
OVA_Uterus	0.724	0.717	0.725	*0.737*	0.697	0.716
parkinson-speech-uci	0.740	0.750	0.742	0.710	0.733	*0.779*
Avg	0.743	0.757	0.759	0.750	0.744	*0.782*

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Table 5. G-means of XGBoost obtained with different methods

	Baseline	SMOTE	Poly-Fit-SMOTE	ProWSyn	SMOTE-IPF	C-EO2
BurkittLymphoma	0.801	0.77	0.757	0.789	0.774	*0.828*
GCM	0.678	0.723	0.73	0.715	0.717	*0.733*
hepatitisC	0.598	0.685	*0.702*	0.683	0.652	0.699
hiva_agnostic	0.611	0.627	*0.634*	0.62	0.589	0.63
lsvt	0.65	*0.702*	0.699	0.683	0.68	0.699
lung-multi	0.788	0.86	0.818	0.868	0.834	*0.901*
micro-mass	0.833	0.851	0.86	*0.879*	0.865	0.862
OVA_Ovary	0.714	0.784	0.78	0.78	0.806	*0.815*
OVA_Uterus	0.658	0.693	0.701	0.722	0.677	*0.706*
parkinson-speech-uci	0.662	0.711	0.725	0.71	0.703	*0.744*
Avg	0.699	0.741	0.741	0.745	0.730	*0.762*

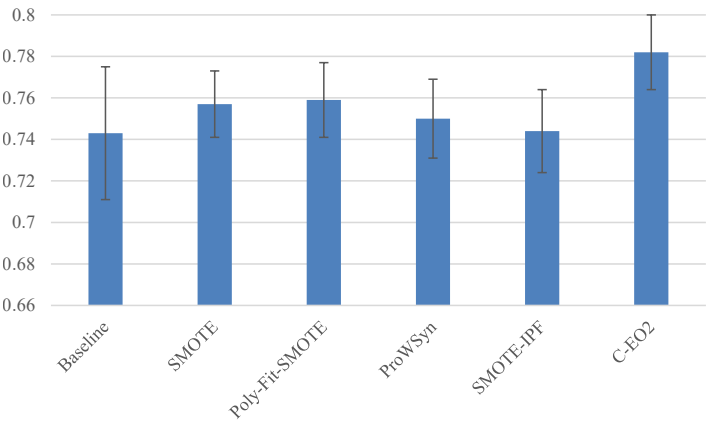


Figure 15. Average AUC rates and their standard deviations of XGBoost by different methods

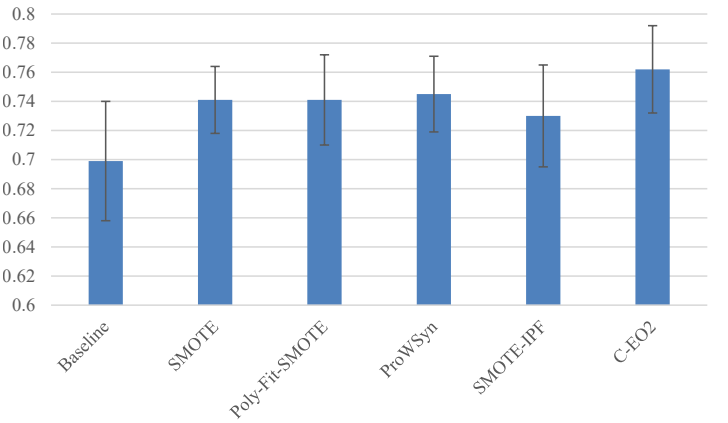


Figure 16. Average G-means and their standard deviations of XGBoost by different methods

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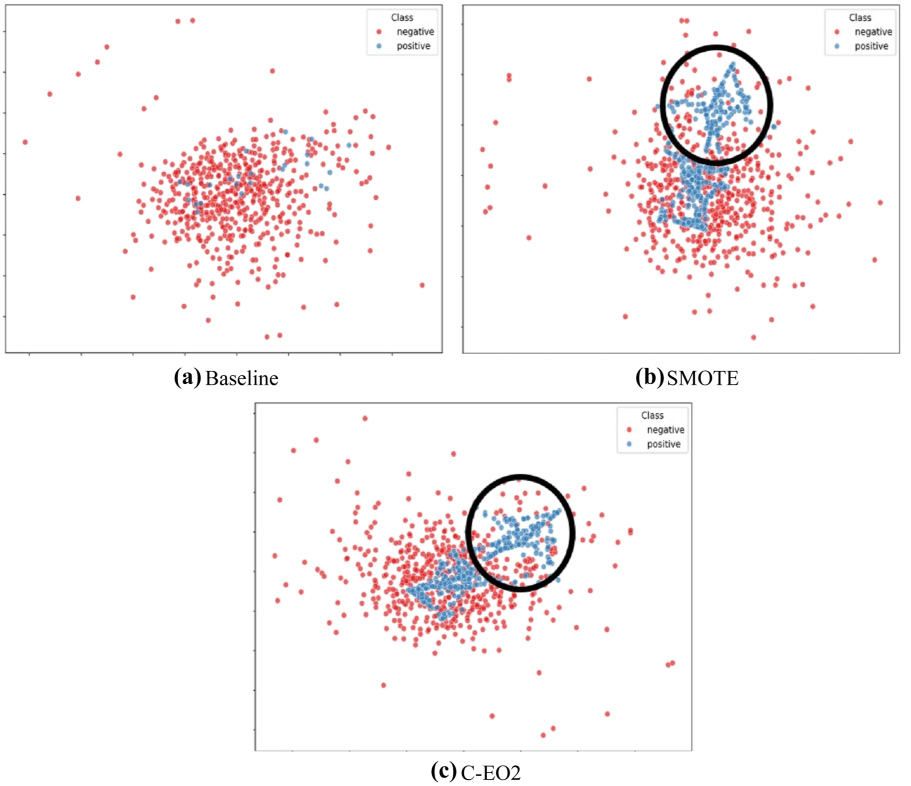


Figure 17. Feature distributions by baseline, SMOTE and C-EO2 methods

C-EO2, with the positive class appearing more compact and well-clustered. These observations highlight not only the practical value of over-sampling but also the enhanced effectiveness of the C-EO2 ensemble method in improving class separability.

4.4.4 Computational complexity analysis. Table 6 presents the computational complexities of the various over-sampling methods using Big O notation. In the case of SMOTE, N denotes the number of minority class samples, and d represents the number of feature dimensions. For Poly-Fit-SMOTE, k refers to the number of nearest neighbors, p to the degree of the polynomial, and M to the number of synthetic samples generated – each requiring a polynomial evaluation. For SMOTE-IPF, I indicates the number of iterations required during the iterative filtering process.

Table 6. Computational complexities of different methods

Algorithms	Computational complexities
SMOTE	$O(N^2d)$
Poly-Fit-SMOTE	$O(N^2d) + O(N\times k\times p^3) + O(M\times p)$
ProWSyn	$O(N^2d) + O(N\log N)$
SMOTE-IPF	$O(N^2d) + O(I\times N)$
C-EO2 (i.e. ProWSyn + SMOTE-IPF)	$O(2N^2d) + O(N\log N) + O(I\times N)$

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Although C-EO2 achieves the highest performance in terms of AUC and G-mean, its computational complexity, stemming from the combined execution of ProWSyn and SMOTE-IPF, is greater than that of any of the three individual over-sampling algorithms. This highlights a clear trade-off between classification performance and computational efficiency, which should be considered when selecting an appropriate over-sampling strategy for practical applications.

5. Conclusion

Data re-sampling is a critical step in data pre-processing for addressing class imbalance, which is essential for constructing more effective classifiers. Among these techniques, over-sampling methods that generate synthetic samples for the minority class have been extensively studied. Although a wide range of over-sampling algorithms has been proposed in the literature, they often produce different synthetic samples for the same imbalanced dataset, making it difficult to determine the most suitable method across various application domains. To address this issue, we propose a series of over-sampling ensemble methods that integrate multiple over-sampling results through parallel and serial combination strategies.

The first experimental study, conducted on 58 binary class imbalanced datasets, demonstrates that the parallel combination approaches outperform both the individual over-sampling methods and the serial ensemble methods. In particular, the ensemble method combining ProWSyn and SMOTE-IPF yields the best classification performance when applied to XGBoost and C4.5 classifiers. The second experimental study, which involves 10 medical domain datasets containing both binary and multi-class problems, further confirms that the proposed method surpasses the four baseline over-sampling algorithms.

Several future directions can be explored to enhance this research. First, feature selection and feature embedding techniques (Sisodia and Sisodia, 2022c) may be integrated to reduce dimensionality in the training data, and their effects on the performance of over-sampling ensembles merit further investigation. Second, other types of re-sampling strategies, such as under-sampling and hybrid sampling, could be incorporated into the proposed parallel and serial frameworks to assess the effectiveness of these ensemble variations. Third, advanced classification techniques, including stacked generalization (Sisodia and Sisodia, 2023a) and transfer learning (Sisodia and Sisodia, 2023b), could be adopted to further enhance model performance when combined with the over-sampling ensemble approaches. Lastly, given the wide variability in class imbalance ratios, feature dimensions and sample sizes across datasets from different domains, future work should explore the adaptability and generalizability of over-sampling ensemble methods to domain-specific challenges.

Notes

1. https://sci2s.ugr.es/keel/datasets.php
2. https://www.openml.org/search?type=data&sort=runs&status=active

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Corresponding author
Wei-Chao Lin can be contacted at: viclin@gap.cgu.edu.tw

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